

Exercises II.1

Connor Mooneyhan

June 26, 2024

1.1. Write a careful proof that every group is the group of isomorphisms of a groupoid. In particular, every group is the group of automorphisms of some object in some category.

Proof. Given a group $(G, \cdot)$, define a category $\mathbf{C}_G$ consisting of exactly one object A , and for each $g \in G$ some morphism $g_{\mathbf{C}} : A \rightarrow A$ such that given any $h \in G$, $h_{\mathbf{C}}g_{\mathbf{C}} := (gh)_{\mathbf{C}}$. Indeed, $e_{\mathbf{C}}$ is the identity morphism since given any $g \in G$,

$$g_{\mathbf{C}}e_{\mathbf{C}} = (eg)_{\mathbf{C}} = g_{\mathbf{C}} = (ge)_{\mathbf{C}} = e_{\mathbf{C}}g_{\mathbf{C}}.$$

Similarly, each morphism $g_{\mathbf{C}}$ is an isomorphism since

$$(g^{-1})_{\mathbf{C}}g_{\mathbf{C}} = (gg^{-1})_{\mathbf{C}} = e_{\mathbf{C}} = (g^{-1}g)_{\mathbf{C}} = g_{\mathbf{C}}(g^{-1})_{\mathbf{C}}.$$

Associativity follows directly from the definitions as well by (given $a, b, c \in G$)

$$(a_{\mathbf{C}}b_{\mathbf{C}})c_{\mathbf{C}} = (ba)_{\mathbf{C}}c_{\mathbf{C}} = (c(ba))_{\mathbf{C}} = ((cb)a)_{\mathbf{C}} = a_{\mathbf{C}}(cb)_{\mathbf{C}} = a_{\mathbf{C}}(b_{\mathbf{C}}c_{\mathbf{C}})$$

Thus $\mathbf{C}_G$ is a groupoid with exactly one object. $\square$

1.2. Consider the ‘sets of numbers’ listed in §1.1, and decide which are made into groups by conventional operations such as $+$ and $\cdot$. Even if the answer is negative (for example, $(\mathbb{R}, \cdot)$ is not a group), see if variations on the definition of these sets lead to groups (for example, $(\mathbb{R}^*, \cdot)$ is a group.)

Proof. Indeed, $(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$, $(\mathbb{C}, +)$ are all groups with 0 as their identity and ‘negatives’ as inverses. The set $(\{-1, 1\}, \cdot)$ is a group, though there seems to be no larger subset of $\mathbb{Z}$ which can be made into a group by multiplication since no other integer has an integer inverse. With $\mathbb{Q}$ and $\mathbb{R}$, though, the sets $\mathbb{Q}^* = \mathbb{Q} \setminus \{0\}$ and $\mathbb{R}^* = \mathbb{R} \setminus \{0\}$ form groups with standard multiplication since they are naturally associative, 1 is the identity, and every nonzero element has a multiplicative inverse. Finally, $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$ is also a group under multiplication since complex multiplication is associative, 1 is the inverse, and each nonzero complex number $a + bi$ has a multiplicative inverse, namely $\frac{a}{a^2+b^2} - \frac{b}{a^2+b^2}i$. $\square$

1.3. Prove that $(gh)^{-1} = h^{-1}g^{-1}$ for all elements g, h of a group G .

Proof. Indeed,

$$(gh)(h^{-1}g^{-1}) = g(hh^{-1})g^{-1} = geg^{-1} = gg^{-1} = e,$$

and any right inverse in a group is also a left inverse, so $h^{-1}g^{-1} = (gh)^{-1}$. $\square$

1.4. Suppose that $g^2 = e$ for all elements g of a group G ; prove that G is commutative.

Proof. If $g^2 = e$ for each $g \in G$, then each element is its own inverse. Applying this fact strategically gives us

$$gh = g^{-1}h^{-1} = (hg)^{-1} = hg.$$

$\square$

1.5. The ‘multiplication table’ of a group is an array compiling the results of all multiplications $g \cdot h$. Prove that every row and every column of the multiplication table of a group contains all elements of the group exactly once (like Sudoku diagrams!).

Proof. For elements $g, h \in G$, the value in row g and column h is defined to be gh . Given two “rows” g_1, g_2 and a “column” h , if $g_1h = g_2h$, then $g_1hh^{-1} = g_2hh^{-1} \implies g_1 = g_2$, so we conclude that within a column, no two (different) rows have the same value, so each row has a unique value. In much the same way, given a row g and two “columns” h_1, h_2 , we have

$$gh_1 = gh_2 \implies g^{-1}gh_1 = g^{-1}gh_2 \implies h_1 = h_2,$$

so each column’s value within a row is unique.

This shows that *if* an element $g \in G$ is in a row or column, it appears in that row or column exactly once, so we now need to show that each g is, in fact, in each column or row. This is easy, since given any $g, h \in G$, $g(g^{-1}h) = h$, so h occupies the $g^{-1}h$ column of row g . Similarly, h is in the hg^{-1} row of column g , so indeed each element appears at least once in each column and each row, and by the previous paragraph, this means it appears *exactly* once in each column and each row. $\square$

1.6. Prove that there is only *one* possible multiplication table for G if G has exactly 1, 2, or 3 elements. Analyze the possible multiplication tables for groups with exactly 4 elements, and show that there are *two* distinct tables, up to reordering the elements of G . Use these tables to prove that all groups with ≤ 4 elements are commutative.

Proof. If there is exactly 1 element in G , it must be the identity, and the product of the identity with itself must be the identity, so indeed the multiplication table is completely determined. This is trivially commutative.

If G is of order 2, call its elements e and a , where e is the identity and a is the other element. Necessarily, $ea = a = ae$ and $ee = e$, so since each element

needs to appear in each row and column exactly once, we must have $aa = e$ since it could not be a again. We can easily see that this makes G commutative.

If G has order 3, with elements e (the identity), a , and b , then we have

	e	a	b
e	e	a	b
a	a	b	e
b	b	e	a

where the identity row and identity column come from the identity rules, then aa must be b because if it were e , there would need to be a second b in the third row and third column. The rest fills itself in using our trick of each element being in each row and column exactly once.

Now, if G has 4 elements, e , a , b , and c , then the following unfinished multiplication table is forced

	e	a	b	c
e	e	a	b	c
a	a			
b	b			
c	c			

and now we have a choice to make: should ab be c or e ? If we choose $ab = e$, then the following diagram is determined:

	e	a	b	c
e	e	a	b	c
a	a	c	e	b
b	b	e	c	a
c	c	b	a	e

(1)

If we choose $ab = c$, then we have

	e	a	b	c
e	e	a	b	c
a	a		c	
b	b	c		
c	c			

This leads us to another choice: should ac be e or b ? If $ac = e$, then we get

	e	a	b	c
e	e	a	b	c
a	a	b	c	e
b	b	c	e	a
c	c	e	a	b

(2)

and if $ac = b$, then

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c		
c	c	b		

so we have one final decision to make: does bc equal a or e ? If $bc = a$, then

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

(3)

and if $bc = e$, then

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	a	e
c	c	b	e	a

(4)

Thus there are four possible multiplication tables, so we must justify the claim that there are in fact *two* up to reordering the elements of G . Here are the four tables presented together:

(1)

	e	a	b	c
e	e	a	b	c
a	a	c	e	b
b	b	e	c	a
c	c	b	a	e

(2)

	e	a	b	c
e	e	a	b	c
a	a	b	c	e
b	b	c	e	a
c	c	e	a	b

(3)

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

(4)

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	a	e
c	c	b	e	a

The two types of multiplication table here are those where there is exactly one element of order 2, namely (1), (2), and (3), and those for whom every non-identity element is of order 2, which is (4). Indeed, we can transform (1) into (2) (up to renaming of non-identity elements) by swapping the b and c columns:

	e	a	b	c
e	e	a	b	c
a	a	c	e	b
b	b	e	c	a
c	c	b	a	e

→

	e	a	c	b
e	e	a	c	b
a	a	c	b	e
c	c	b	e	a
b	b	e	a	c

and it can be seen easily that replacing b with c and c with b in the latter table yields table (2) exactly.

We can perform a similar trick to transform (1) into (4), this time swapping a and c , like so

	e	a	b	c
e	e	a	b	c
a	a	c	e	b
b	b	e	c	a
c	c	b	a	e

 $\rightarrow$

	e	c	b	a
e	e	c	b	a
c	c	e	a	b
b	b	a	c	e
a	a	b	e	c

where then swapping out c for a and a for c gives (4) exactly.

In this way, there is a sort of “fundamental structure” preserved through (1), (2), and (4), and a different structure in (3). It is also clear that all of these are commutative, since they are symmetric about the NW-SE “diagonal”. $\square$

1.7. Prove Corollary 1.11.

Corollary 1.11: Let g be an element of finite order, and let $N \in \mathbb{Z}$. Then

$$g^N = e \iff N \text{ is a multiple of } |g|.$$

Proof.

($\Leftarrow$) If N is a multiple of $|g|$, then for some $m \in \mathbb{Z}$,

$$g^N = g^{m|g|} = (g^{|g|})^m = e^m = e.$$

($\Rightarrow$) If $N > 0$, then this direction is simply a restatement of Lemma 1.10. If $N = 0$, then of course N is a multiple of $|g|$ because 0 is a multiple of every integer. If $N < 0$, then

$$g^{-N} = (g^N)^{-1} = e^{-1} = e,$$

so $-N = m|g|$ for some $m \in \mathbb{Z}$ by Lemma 1.10, and thus $N = -m|g|$, making N an integer multiple of $|g|$. $\square$

1.8. Let G be a finite abelian group with exactly one element f of order 2. Prove that $\prod_{g \in G} g = f$.

Proof. We begin by observing that since there is an element of order 2, $|G|$ is even. We can then write G as $\{e, f, g_1, \dots, g_{2n}\}$ where $2n = |G| - 2$. If $|G| = 2$, then obviously $ef = f$, so the theorem holds. Thus we can assume that $n > 0$.

Important to note is the fact that each g_j is not its own inverse (if it were, it would have order ≤ 2). Furthermore, f and e are their own inverses, so they cannot be the inverse of any g_j . Thus for each g_j , there is exactly one $k \neq j$ where $1 \leq k \leq 2n$ for which $g_k = (g_j)^{-1}$. We can then “pair up” the g_j s such that, after potential re-indexing, we get

$$\prod_{g \in G} g = ef g_1 g_1^{-1} \cdots g_n g_n^{-1} = fe^n = f.$$

$\square$

1.9. Let G be a finite group, of order n , and let m be the number of elements $g \in G$ of order exactly 2. Prove that $n - m$ is odd. Deduce that if n is even, then G necessarily contains elements of order 2.

Proof. This proof will use a strategy similar to the previous one. The identity element $e \in G$ is of order 1, so we can express G as $\{e, f_1, \dots, f_m, g_1, \dots, g_{n-m-1}\}$ where $f_1, \dots, f_m \in G$ are all the elements of order exactly 2, and $g_1, \dots, g_{n-m-1}$ are the elements of order more than 2. As mentioned in the proof of the previous theorem, each g_j is not its own inverse, and each of $e, f_1, \dots, f_m$ is its own inverse, so for each g_j , there must exist (exactly) one $k \neq j$ such that $g_k = (g_j)^{-1}$.

Let $\hat{G} = \{g_1, \dots, g_{n-m-1}\}$. We define a relation $\sim$ on $\hat{G}$ like so: given $a, b \in \hat{G}$, $a \sim b$ if and only if either $a = b$ or $a^{-1} = b$. This is reflexive by definition, and symmetric since if $a \sim b$ and $a \neq b$, then $a^{-1} = b$, so $a = b^{-1}$ and $b \sim a$. Given $a \sim b$ and $b \sim c$, if either $a = b$ or $b = c$, transitivity is evident. Assuming, then, that $a \neq b \neq c$, we have $a^{-1} = b$ and $b^{-1} = c$, so $c = b^{-1} = (a^{-1})^{-1} = a$. Thus $a \sim c$ and we've shown that $\sim$ is an equivalence relation.

We can now quotient out by $\sim$ and study the members of $\hat{G}/\sim$. Each $S \in \hat{G}/\sim$ contains exactly 2 elements since, as seen in the proof of transitivity, if there is a third element (or more), it is equal to one of the other elements. Thus each $S \in \hat{G}/\sim$ has at most 2 elements, and of course it also has at least 2 elements because each element of $\hat{G}$ has an inverse in $\hat{G}$, as observed above. Since $\hat{G}/\sim$ is a partition of $\hat{G}$ and each $S \in \hat{G}/\sim$ has cardinality 2,

$$\text{card}(\hat{G}) = \text{card}\left(\bigcup \hat{G}/\sim\right) = 2 * \text{card}(\hat{G}/\sim).$$

Thus $\text{card}(\hat{G}) = n - m - 1$ is even, so $n - m$ is odd as desired.

We conclude, as instructed, that if n is even, there must be elements of order 2 in G since if there were none, m as defined in this exercise would be 0 and thus $n - m = n$ would be even. $\square$

1.10. Suppose the order of g is odd. What can you say about the order of g^2 ?

Proof. By Proposition 1.13, g^2 also has finite order, and in particular,

$$|g^2| = \frac{|g|}{\gcd(2, |g|)} = \frac{|g|}{1} = |g|.$$

$\square$

1.11. Prove that for all g, h in a group G , $|gh| = |hg|$. (Hint: Prove that $|aga^{-1}| = |g|$ for all a, g in G .)

Proof. First, we show that given $g, h \in G$, $|hgh^{-1}| = |g|$. We will do this by showing that $(hgh^{-1})^m = e$ if and only if $g^m = e$ for any integer m . Observe that

$$(hgh^{-1})^n = \underbrace{(hgh^{-1})(hgh^{-1}) \cdots (hgh^{-1})}_n = hg^n h^{-1}$$

since, after rearranging parentheses, each $h^{-1}h = e$ “vanishes.” Thus if $(hgh^{-1})^m = e$ for some m , then

$$\begin{aligned}(hgh^{-1})^m &= hg^mh^{-1} = e \\ \implies g^m &= h^{-1}h = e\end{aligned}$$

where the implication comes by multiplying on the left by h^{-1} and on the right by h . Conversely, if $g^m = e$, then clearly

$$(hgh^{-1})^m = hg^mh^{-1} = heh^{-1} = hh^{-1} = e.$$

□

1.12. In the group of invertible 2×2 matrices, consider

$$g = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad h = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}.$$

Verify that $|g| = 4$, $|h| = 3$, and $|gh| = \infty$.

Proof. First we calculate

$$\begin{aligned}g^4 &= \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},\end{aligned}$$

so $g^4 = e$, and since we did not reach the identity matrix until the last step, $g^m \neq e$ for any $m < 4$. Thus $|g| = 4$.

Also,

$$\begin{aligned}h^3 &= \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix} \\ &= \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},\end{aligned}$$

so, following the same reasoning as before, $|h| = 3$.

On the other hand,

$$gh = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

and for any positive integer m , it is easy to see that

$$(gh)^m = \begin{pmatrix} 1 & m \\ 0 & 1 \end{pmatrix},$$

so indeed $|gh| = \infty$. $\square$

1.13. Give an example showing that $|gh|$ is not necessarily equal to $\text{lcm}(|g|, |h|)$, even if g and h commute.

Proof. Let f be an element of order 2 in a group G . Then $|ff| = |e| = 1 \neq 2 = \text{lcm}(|f|, |f|)$. $\square$

1.14. As a counterpoint to Exercise 1.13, prove that if g and h commute and $\text{gcd}(|g|, |h|) = 1$, then $|gh| = |g||h|$. (Hint: Let $N = |gh|$; then $g^N = (h^{-1})^N$. What can you say about this element?)

Proof. Observe that $e = gh^{|gh||h|} = g^{|gh||h|}h^{|gh||h|} = g^{|gh||h|}$. Thus $|g|$ divides $|gh||h|$, so since $\text{gcd}(|g|, |h|) = 1$, $|g|$ divides $|gh|$. We can show similarly that $|h|$ divides $|gh|$ by swapping g and h in the above calculation. Thus, again since $\text{gcd}(|g|, |h|) = 1$, $|g||h|$ divides $|gh|$. We already know that $|gh|$ divides $|g||h|$ by Proposition 1.14, so we conclude that $|gh| = |g||h|$. $\square$

1.15. Let G be a commutative group, and let $g \in G$ be an element of maximal finite order, that is, such that if $h \in G$ has finite order, then $|h| \leq |g|$. Prove that in fact if h has finite order in G , then $|h|$ divides $|g|$. (Hint: Argue by contradiction. If $|h|$ is finite but does not divide $|g|$, then there is a prime integer p such that $|g| = p^m r$, $|h| = p^n s$, with r and s relatively prime to p and $m < n$. Use Exercise 1.14 to compute the order of $g^{p^m} h^s$.)

Proof. Let $g \in G$ be of maximal finite order and $h \in G$ of finite order. If $|h|$ does not divide $|g|$, then $|g| = p^m r$ and $|h| = p^n s$, where r and s are relatively prime to p and $m < n$. Then $r = a|g^{p^m}|$ for some positive integer a by Corollary 1.11, so $g^{p^m \frac{r}{a}} = e$, implying that $a = 1$. Thus $|g^{p^m}| = r$, and by similar logic, $|h^s| = p^n$. In particular, this tells us that $|g^{p^m}|$ and $|h^s|$ are relatively prime. We conclude, by Exercise 1.14, that

$$|g^{p^m} h^s| = |g^{p^m}| |h^s| = rp^n > |g|$$

contradicting the hypothesis that g is of maximal finite order. $\square$